

ACTIVITY 22 — LEARNER WORKSHEET

SPC Control Chart and Statistical Process Analysis

TYPE	DURATION	MAPS TO	TOPIC
Workshop	60 minutes	LO5	5 - Monitor and Control the Project

What you are doing

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

What you will produce: control chart, stability verdict, capability summary — submitted to artifacts/.

What you need before you start

- Lab 21 corrective actions and their implementation date
- Lab 06 success criterion SC-1
- Lab 20 trend analysis method

Your steps

1. Take the measurement data. Contoso measures median registration completion time every working day. Charter success criterion SC-1 requires the completed system to achieve under 4.0 minutes, down from the 11-minute legacy baseline.
2. Compute the centre line and control limits. This is an individuals chart (X-mR), used when each measurement is a single value rather than a subgroup average. The control limits come from the moving range, not from the standard deviation.
3. Plot the chart. Enter the full 20-point series in order:
4. Apply the out-of-control rules. A single point outside the limits is only one of several signals. Apply the full set - these are the Western Electric / Nelson rules, and the exam expects the rule of seven in particular.
5. Separate the causes and decide the response
6. Assess capability against the charter target. Now bring in the specification limit, which has been absent so far and deliberately so.
7. Write the quality verdict and actions. Create artifacts/22-stability-verdict.md:
8. Answer the exam-style scenarios

Your working space

Complete every field. Keep this worksheet — it is your reference for the rest of the course.

Step 1 — Take the measurement data

Step 2 — Compute the centre line and control limits

Step 3 — Plot the chart

Step 4 — Apply the out-of-control rules

Step 5 — Separate the causes and decide the response

Step 6 — Assess capability against the charter target

Step 7 — Write the quality verdict and actions

Step 8 — Answer the exam-style scenarios

Deliverable checklist

- 22-control-chart.md - the centre line, moving ranges, estimated sigma, and both the individuals and moving-range control limits, all computed by hand with the working shown.
- A statement of why limits were computed from days 1-15 rather than all 20, and what happens to the limits if you use all 20.
- A screenshot or export of your SPC tool output showing the 20 plotted points against limits derived from the stable period.
- The out-of-control rules table with each rule assessed, including the explicit rule-of-seven run analysis on days 1-15.

- 22-stability-verdict.md - the common cause / assignable cause separation, the identified assignable cause with its timing and magnitude evidence, and the correct response to each.
- 22-capability-summary.md - the Statistics tool output, the Cpk calculation, and the interpretation against the capability bands.
- A written statement of the difference between the stability finding and the capability finding, and why fixing only the day-16 shift would leave SC-1 unmet.
- The four quality actions with expected recovery figures and the honest post-action Cpk forecast.
- Written answers to the three scenarios.